

A Self-Contained Proof of Sharpness for Nearest-Neighbor Bond Percolation on $\mathbb{Z}^d$

Abstract

This note gives a self-contained version of the Duminil-Copin–Tassion sharpness argument, specialized to nearest-neighbor Bernoulli bond percolation on $\mathbb{Z}^d$, $d \geq 2$, and written with the open-edge parameter $p \in [0, 1]$. We prove only the sharpness statements: exponential decay below criticality and a quantitative lower bound above criticality. The susceptibility estimate is intentionally omitted.

1 Model and statement

Let $d \geq 2$. The vertex set is $\mathbb{Z}^d$, and two vertices x, y are adjacent, written $x \sim y$, if $\|x - y\|_1 = 1$. In nearest-neighbor bond percolation with parameter $p \in [0, 1]$, each edge is declared open with probability p , independently of all other edges. The resulting product measure is denoted by $\mathbb{P}_p$.

For $S \subseteq \mathbb{Z}^d$, write

$$x \overset{S}{\longleftrightarrow} y$$

if x and y are connected by an open path all of whose vertices lie in S . If $S = \mathbb{Z}^d$, we simply write $x \longleftrightarrow y$.

Let

$$\Lambda_n := \{x \in \mathbb{Z}^d : \|x\|_1 \leq n\}, \quad \theta(p) := \mathbb{P}_p(0 \longleftrightarrow \infty),$$

where

$$\{0 \longleftrightarrow \infty\} := \bigcap_{n \geq 1} \{0 \longleftrightarrow \Lambda_n^c\}.$$

Define

$$p_c := \inf\{p \in [0, 1] : \theta(p) > 0\}.$$

Theorem 1 (Sharpness on $\mathbb{Z}^d$). *For nearest-neighbor bond percolation on $\mathbb{Z}^d$, $d \geq 2$, the following hold.*

(i) *If $p < p_c$, then there exists $c = c(p) > 0$ such that, for every $n \geq 0$,*

$$\mathbb{P}_p(0 \longleftrightarrow \Lambda_n^c) \leq e^{-cn}.$$

(ii) *If $p_c < p < 1$, then*

$$\theta(p) \geq \frac{p - p_c}{p(1 - p_c)}.$$

The endpoint $p = 1$ is trivial, since $\theta(1) = 1$.

The proof proceeds through an auxiliary critical point defined in terms of finite sets. For a finite set $S \subset \mathbb{Z}^d$ with $0 \in S$, define

$$\partial_E S := \{(x, y) : x \in S, y \notin S, x \sim y\},$$

where boundary edges are oriented from S to its complement, and set

$$\varphi_p(S) := p \sum_{(x,y) \in \partial_E S} \mathbb{P}_p(0 \xleftrightarrow{S} x).$$

Define

$$\tilde{p}_c := \sup\{p \in [0, 1] : \text{there exists a finite } S \ni 0 \text{ such that } \varphi_p(S) < 1\}.$$

Because $\varphi_p(S)$ is nondecreasing in p , the definition implies:

$$p < \tilde{p}_c \implies \text{there exists finite } S \ni 0 \text{ with } \varphi_p(S) < 1, \quad (1)$$

$$p > \tilde{p}_c \implies \varphi_p(S) \geq 1 \text{ for every finite } S \ni 0. \quad (2)$$

Indeed, let

$$\mathcal{I} := \{r \in [0, 1] : \text{there exists a finite } S \ni 0 \text{ such that } \varphi_r(S) < 1\}.$$

For the first implication, if $p < \tilde{p}_c = \sup \mathcal{I}$, then there is some $q \in \mathcal{I}$ with $q > p$. Choose a finite $S \ni 0$ such that $\varphi_q(S) < 1$; by monotonicity in the edge parameter,

$$\varphi_p(S) \leq \varphi_q(S) < 1.$$

For the second implication, if $p > \tilde{p}_c$ and some finite $S \ni 0$ satisfied $\varphi_p(S) < 1$, then $p \in \mathcal{I}$, contradicting $p > \sup \mathcal{I}$.

We prove the theorem with $\tilde{p}_c$ in place of p_c , and then show $\tilde{p}_c = p_c$.

2 Russo's formula in the form used below

Let A be an increasing event depending on finitely many edges. For an edge e , call e *closed-pivotal* for A if e is closed, A does not occur, and forcing e to be open makes A occur.

Lemma 1 (Russo formula, closed-pivotal form). *If A is increasing and depends on finitely many edges, then*

$$\frac{d}{dp} \mathbb{P}_p(A) = \frac{1}{1-p} \sum_e \mathbb{P}_p(e \text{ is closed-pivotal for } A), \quad 0 < p < 1.$$

The sum is over the finite set of edges on which A depends.

Proof. Let F be the finite edge set on which A depends. For $e \in F$, let B_e be the event, depending only on the states of edges in $F \setminus \{e\}$, that changing the state of e changes whether A occurs. For increasing A , this means that A occurs when e is open and fails when e is closed. Hence

$$\frac{\partial}{\partial p_e} \mathbb{P}(A) = \mathbb{P}(B_e).$$

In the homogeneous model $p_e = p$ for all e , so

$$\frac{d}{dp} \mathbb{P}_p(A) = \sum_{e \in F} \mathbb{P}_p(B_e).$$

Since B_e is independent of the state of e ,

$$\mathbb{P}_p(e \text{ is closed-pivotal}) = (1-p) \mathbb{P}_p(B_e).$$

Substituting gives the claimed formula. □

3 The fundamental differential inequality

For a finite set $\Lambda \subset \mathbb{Z}^d$ with $0 \in \Lambda$, define

$$A_\Lambda := \{0 \longleftrightarrow \Lambda^c\}, \quad f_\Lambda(p) := \mathbb{P}_p(A_\Lambda).$$

The event A_Λ depends on finitely many edges. Indeed, since Λ is finite and the model is nearest-neighbor,

$$A_\Lambda = \bigcup_{(x,y) \in \partial_E \Lambda} \{0 \xrightarrow{\Lambda} x\} \cap \{xy \text{ is open}\}.$$

The right-hand side only uses the finitely many edges with both endpoints in Λ and the finitely many boundary edges in $\partial_E \Lambda$.

Lemma 2 (Differential inequality). *For every finite $\Lambda \ni 0$ and every $0 < p < 1$,*

$$f'_\Lambda(p) \geq \frac{1}{p(1-p)} \inf_{\substack{S \subset \Lambda \\ 0 \in S}} \varphi_p(S) (1 - f_\Lambda(p)).$$

Proof. Let

$$\mathcal{S} := \{x \in \Lambda : x \not\longleftrightarrow \Lambda^c\}.$$

Thus $0 \in \mathcal{S}$ exactly on the event A_Λ^c .

By the closed-pivotal Russo formula,

$$f'_\Lambda(p) = \frac{1}{1-p} \sum_e \mathbb{P}_p(e \text{ is closed-pivotal for } A_\Lambda).$$

We decompose according to the value of $\mathcal{S}$. Fix a set $S \subset \Lambda$ with $0 \in S$ and an oriented boundary edge $(x, y) \in \partial_E S$. On the event $\{\mathcal{S} = S\}$, the edge xy is closed-pivotal for A_Λ if and only if

$$0 \xrightarrow{S} x.$$

Indeed, assume first that $0 \xrightarrow{S} x$ and $\mathcal{S} = S$. If $y \in \Lambda \setminus S$, then y is connected to Λ^c by the definition of $\mathcal{S}$; if $y \notin \Lambda$, then y is connected to Λ^c by the trivial path. The edge xy must be closed, since otherwise $x \in S$ would be connected to Λ^c . Opening xy therefore connects 0 to Λ^c , while A_Λ does not occur on $\{\mathcal{S} = S\}$ because $0 \in S$. Conversely, suppose that xy is closed-pivotal and $\mathcal{S} = S$. After xy is opened, every open path from 0 to Λ^c must use the edge xy , since A_Λ failed before it was opened. The endpoint of xy connected to 0 before the opening cannot be y : if $y \in \Lambda \setminus S$, then $y \longleftrightarrow \Lambda^c$, and if $y \notin \Lambda$, then $y \in \Lambda^c$. Hence 0 is connected to x before the opening. This connection cannot visit $\Lambda \setminus S$ or Λ^c , since every vertex of $\Lambda \setminus S$ is connected to Λ^c and visiting Λ^c would already give A_Λ . Thus $0 \xrightarrow{S} x$.

We next give a precise measurability statement. For fixed S , let ω^{out} be the configuration obtained from ω by closing all edges with both endpoints in S . Then $\{\mathcal{S} = S\}$ is equivalent to the following two conditions, both evaluated in ω^{out} :

- (a) no vertex of S is connected to Λ^c ;
- (b) every vertex of $\Lambda \setminus S$ is connected to Λ^c .

The implication from these two conditions to $\mathcal{S} = S$ follows by a last-exit argument: if, after restoring the internal edges of S , some vertex of S were connected to Λ^c , then along such a path take the last edge leaving S ; the S -endpoint of this edge would already be connected to Λ^c without using any edge internal to S , contradicting (a). Conversely, if $\mathcal{S} = S$ in the full configuration, then (a) is immediate after deleting edges. For (b), take $z \in \Lambda \setminus S$. Since $z \longleftrightarrow \Lambda^c$ in the full configuration, choose an open path from z to Λ^c . If no such path existed in ω^{out} , the path would have to use an internal edge of S ; taking its last exit from S would again show that some vertex of S is connected to Λ^c in ω^{out} , contradicting $\mathcal{S} = S$. Hence $\{\mathcal{S} = S\}$ is measurable with respect to the edges that are not internal to S .

The event $\{0 \xleftrightarrow{S} x\}$ depends only on edges internal to S , so it is independent of $\{\mathcal{S} = S\}$. Therefore

$$\mathbb{P}_p(0 \xleftrightarrow{S} x, \mathcal{S} = S) = \mathbb{P}_p(0 \xleftrightarrow{S} x) \mathbb{P}_p(\mathcal{S} = S).$$

Using only the closed-pivotal edges oriented from S to S^c , we get

$$\begin{aligned} f'_\Lambda(p) &\geq \frac{1}{1-p} \sum_{\substack{S \subset \Lambda \\ 0 \in S}} \sum_{(x,y) \in \partial_E S} \mathbb{P}_p(0 \xleftrightarrow{S} x, \mathcal{S} = S) \\ &= \frac{1}{1-p} \sum_{\substack{S \subset \Lambda \\ 0 \in S}} \sum_{(x,y) \in \partial_E S} \mathbb{P}_p(0 \xleftrightarrow{S} x) \mathbb{P}_p(\mathcal{S} = S) \\ &= \frac{1}{p(1-p)} \sum_{\substack{S \subset \Lambda \\ 0 \in S}} \varphi_p(S) \mathbb{P}_p(\mathcal{S} = S) \\ &\geq \frac{1}{p(1-p)} \inf_{\substack{S \subset \Lambda \\ 0 \in S}} \varphi_p(S) \sum_{\substack{S \subset \Lambda \\ 0 \in S}} \mathbb{P}_p(\mathcal{S} = S). \end{aligned}$$

Finally,

$$\sum_{\substack{S \subset \Lambda \\ 0 \in S}} \mathbb{P}_p(\mathcal{S} = S) = \mathbb{P}_p(0 \in \mathcal{S}) = 1 - f_\Lambda(p).$$

This proves the lemma. □

4 Supercritical lower bound

Lemma 3 (Lower bound above $\tilde{p}_c$). *If $\tilde{p}_c < p < 1$, then*

$$\theta(p) \geq \frac{p - \tilde{p}_c}{p(1 - \tilde{p}_c)}.$$

If $\tilde{p}_c < 1$, the endpoint $p = 1$ also follows by taking $p \uparrow 1$, or directly from $\theta(1) = 1$.

Proof. Fix a finite $\Lambda \ni 0$. If $q > \tilde{p}_c$, then (2) gives

$$\inf_{\substack{S \subset \Lambda \\ 0 \in S}} \varphi_q(S) \geq 1.$$

Thus, for $q \in (\tilde{p}_c, 1)$,

$$f'_\Lambda(q) \geq \frac{1}{q(1-q)}(1 - f_\Lambda(q)).$$

For $q < 1$ we have $f_\Lambda(q) < 1$: closing all edges in $\partial_E \Lambda$ prevents A_Λ , and this has positive probability. Hence $\log(1 - f_\Lambda(q))$ is well-defined, and the preceding inequality gives

$$\frac{d}{dq} \log(1 - f_\Lambda(q)) \leq -\frac{1}{q(1-q)}.$$

Let $\tilde{p}_c < q < p < 1$. Integrating from q to p gives

$$\log \frac{1 - f_\Lambda(p)}{1 - f_\Lambda(q)} \leq - \int_q^p \frac{dt}{t(1-t)} = - \log \frac{p(1-q)}{q(1-p)}.$$

Hence

$$1 - f_\Lambda(p) \leq (1 - f_\Lambda(q)) \frac{q(1-p)}{p(1-q)} \leq \frac{q(1-p)}{p(1-q)}.$$

Letting $q \downarrow \tilde{p}_c$ yields

$$f_\Lambda(p) \geq 1 - \frac{\tilde{p}_c(1-p)}{p(1-\tilde{p}_c)} = \frac{p - \tilde{p}_c}{p(1-\tilde{p}_c)}.$$

Now take $\Lambda = \Lambda_n$ and let $n \rightarrow \infty$. The events $\{0 \longleftrightarrow \Lambda_n^c\}$ decrease to $\{0 \longleftrightarrow \infty\}$, so by continuity from above,

$$\theta(p) = \lim_{n \rightarrow \infty} f_{\Lambda_n}(p) \geq \frac{p - \tilde{p}_c}{p(1-\tilde{p}_c)}.$$

□

5 A boundary inequality

The following lemma is the finite-set expansion estimate that drives the subcritical part.

Lemma 4 (Boundary inequality). *Let $S \subset \mathbb{Z}^d$ be finite, let $u \in S$, and let $B \subset \mathbb{Z}^d$ satisfy $B \cap S = \emptyset$. Then*

$$\mathbb{P}_p(u \longleftrightarrow B) \leq \sum_{(x,y) \in \partial_E S} p \mathbb{P}_p(u \xrightarrow{S} x) \mathbb{P}_p(y \longleftrightarrow B).$$

Here $y \longleftrightarrow B$ means that y is connected to at least one vertex of B , with the convention that the event holds if $y \in B$.

Proof. Let $C = C_S(u)$ be the open cluster of u inside S :

$$C_S(u) := \{x \in S : u \xrightarrow{S} x\}.$$

Fix a possible value $C_0 \subset S$ with $u \in C_0$. On the event $\{C = C_0\}$, if $u \longleftrightarrow B$, choose a self-avoiding open path from u to B . Since $B \cap S = \emptyset$, the path eventually leaves C_0 . Let x be the last vertex of the path lying in C_0 , and let y be the next vertex. Then $x \in C_0$, $x \sim y$, and $y \notin S$; if $y \in S$, the open edge xy would put y in the S -cluster of u , contradicting the choice of C_0 . The remaining part of the path connects y to B while avoiding C_0 .

Therefore

$$\{u \longleftrightarrow B, C = C_0\} \subseteq \bigcup_{\substack{x \in C_0, y \notin S \\ x \sim y}} \{C = C_0\} \cap \{xy \text{ open}\} \cap \{y \xrightarrow{\mathbb{Z}^d \setminus C_0} B\}.$$

For fixed C_0, x, y , these three events are measurable with respect to pairwise disjoint edge sets. More explicitly, $\{C = C_0\}$ only uses edges with both endpoints in S and at least one endpoint in

C_0 ; the event $\{xy \text{ open}\}$ uses the single edge with $x \in C_0$ and $y \notin S$; and $\{y \xleftrightarrow{\mathbb{Z}^d \setminus C_0} B\}$ only uses edges with both endpoints in $\mathbb{Z}^d \setminus C_0$. Hence the three events are independent. By a union bound,

$$\begin{aligned} \mathbb{P}_p(u \longleftrightarrow B, C = C_0) &\leq \sum_{\substack{x \in C_0, y \notin S \\ x \sim y}} \mathbb{P}_p(C = C_0) p \mathbb{P}_p(y \xleftrightarrow{\mathbb{Z}^d \setminus C_0} B) \\ &\leq \sum_{\substack{x \in C_0, y \notin S \\ x \sim y}} \mathbb{P}_p(C = C_0) p \mathbb{P}_p(y \longleftrightarrow B). \end{aligned}$$

Summing over all possible C_0 gives

$$\begin{aligned} \mathbb{P}_p(u \longleftrightarrow B) &\leq \sum_{C_0} \sum_{\substack{x \in C_0, y \notin S \\ x \sim y}} \mathbb{P}_p(C = C_0) p \mathbb{P}_p(y \longleftrightarrow B) \\ &= \sum_{(x,y) \in \partial_E S} p \mathbb{P}_p(x \in C_S(u)) \mathbb{P}_p(y \longleftrightarrow B) \\ &= \sum_{(x,y) \in \partial_E S} p \mathbb{P}_p(u \xleftrightarrow{S} x) \mathbb{P}_p(y \longleftrightarrow B). \end{aligned}$$

□

6 Subcritical exponential decay

Lemma 5 (Exponential decay below $\tilde{p}_c$). *If $p < \tilde{p}_c$, then there exists $c = c(p) > 0$ such that, for every $n \geq 0$,*

$$\mathbb{P}_p(0 \longleftrightarrow \Lambda_n^c) \leq e^{-cn}.$$

Proof. If $p = 0$, the result is immediate. Assume $0 < p < \tilde{p}_c$. By (1), choose a finite set $S \ni 0$ such that

$$\rho := \varphi_p(S) < 1.$$

Choose $L \geq 1$ such that $S \subset \Lambda_{L-1}$. Then every boundary endpoint y appearing in $\partial_E S$ belongs to Λ_L .

Set

$$a_n := \mathbb{P}_p(0 \longleftrightarrow \Lambda_n^c).$$

For $n \geq L$, apply the boundary inequality with $u = 0$ and $B = \Lambda_n^c$:

$$a_n \leq \sum_{(x,y) \in \partial_E S} p \mathbb{P}_p(0 \xleftrightarrow{S} x) \mathbb{P}_p(y \longleftrightarrow \Lambda_n^c).$$

For each boundary endpoint $y \in \Lambda_L$, the event $\{y \longleftrightarrow \Lambda_n^c\}$ implies, after translating y to the origin, the event $\{0 \longleftrightarrow \Lambda_{n-L}^c\}$. Indeed, if y is connected to some $z \in \Lambda_n^c$, then

$$\|z - y\|_1 \geq \|z\|_1 - \|y\|_1 > n - L,$$

so the translated endpoint $z - y$ lies in Λ_{n-L}^c . Hence

$$\mathbb{P}_p(y \longleftrightarrow \Lambda_n^c) \leq a_{n-L}.$$

Consequently,

$$a_n \leq \rho a_{n-L}, \quad n \geq L.$$

Iterating gives

$$a_n \leq \rho^{\lfloor n/L \rfloor}, \quad n \geq 0,$$

where we use the trivial bound $a_m \leq 1$ for the final remainder. Since $\rho < 1$,

$$a_n \leq \exp\left(-\frac{-\log \rho}{2L}n\right) \quad \text{for all } n \geq L.$$

For the finitely many $1 \leq n < L$, we also have $a_n < 1$: closing all edges crossing from Λ_n to Λ_n^c prevents 0 from connecting to Λ_n^c , and this has positive probability because $p < 1$. Therefore, with the convention that the minimum over an empty set is $+\infty$, define

$$c := \min \left\{ \frac{-\log \rho}{2L}, \min_{1 \leq n < L} \frac{-\log a_n}{n} \right\} > 0.$$

Then $a_n \leq e^{-cn}$ for every $n \geq 1$, while the case $n = 0$ is trivial because $a_0 \leq 1 = e^0$. $\square$

7 Identification of the critical point

We now prove $\tilde{p}_c = p_c$. Recall first that θ is nondecreasing in p by the standard product coupling, and that $\theta(1) = 1$.

First, suppose $\tilde{p}_c < 1$. For every $p \in (\tilde{p}_c, 1)$, the supercritical lemma gives $\theta(p) > 0$, hence $p_c \leq p$. Letting $p \downarrow \tilde{p}_c$ gives $p_c \leq \tilde{p}_c$. If $\tilde{p}_c = 1$, then the same inequality is trivial because $p_c \leq 1 = \tilde{p}_c$.

Second, if $p < \tilde{p}_c$, the exponential decay lemma gives

$$\mathbb{P}_p(0 \longleftrightarrow \Lambda_n^c) \leq e^{-cn} \xrightarrow{n \rightarrow \infty} 0.$$

Since the events $\{0 \longleftrightarrow \Lambda_n^c\}$ decrease to $\{0 \longleftrightarrow \infty\}$,

$$\theta(p) = 0.$$

Thus no parameter below $\tilde{p}_c$ belongs to the set $\{p : \theta(p) > 0\}$. Since this set is nonempty, because $\theta(1) = 1$, its infimum is at least $\tilde{p}_c$; that is,

$$\tilde{p}_c \leq p_c.$$

Combining the two inequalities yields

$$\tilde{p}_c = p_c.$$

The supercritical and subcritical lemmas can now be rewritten with p_c in place of $\tilde{p}_c$. This proves sharpness of the phase transition.

Remark 1. *The proof above deliberately omits the susceptibility estimate. The usual Duminil-Copin–Tassion argument also proves finiteness of $\sum_x \mathbb{P}_p(0 \longleftrightarrow x)$ for $p < p_c$; here that step is unnecessary because the boundary inequality already gives the stronger distance-to-boundary exponential decay in the nearest-neighbor case.*

References

- [1] H. Duminil-Copin and V. Tassion, *A new proof of the sharpness of the phase transition for Bernoulli percolation and the Ising model*, Commun. Math. Phys. 343, 725–745 (2016), arXiv:1502.03050.